

Week 08: Monday, AST 5011: Astrophysical Systems

The Galaxy Population

Michael Coughlin cough052@umn.edu

With contributions from Frank van den Bosch (Yale) and Benedikt Diemer (UMD).

Textbook reference: *Introduction to Galaxy Formation and Evolution* by Cimatti, Fraternali, and Nipoti (CFN).

Cosmological Tools

Throughout the course, we will need cosmological calculations such as distances. We use the [Colossus](#) code for this purpose.

The Colossus cosmology module provides standard calculations: densities, distances, and times. Redshift is always the argument.

Cosmological Distances

Different distance measures diverge at high redshift. The luminosity distance d_L grows faster than comoving distance d_C , while the angular diameter distance d_A turns over around $z \sim 1$.

The SDSS Galaxy Sample

The [Sloan Digital Sky Survey](#) (SDSS) provides the largest spectroscopic galaxy sample, with photometry in five bands (*ugriz*) and redshifts for hundreds of thousands of galaxies. We load a pre-processed version of the SDSS DR8 spectroscopic galaxy sample.

Distribution on the Sky

Let's plot the SDSS footprint using a Hammer projection.

Distribution in Redshift

The SDSS spectroscopic sample covers roughly $0 < z < 0.3$, with a tail to higher redshifts.

Large-Scale Structure

A redshift cone plot reveals filaments, voids, and walls in the galaxy distribution. The "Sloan Great Wall" is visible as a massive filament.

What Do SDSS Galaxies Look Like?

Let's plot a collage of SDSS galaxy images from a narrow distance range.

Magnitudes, Extinction, and K-corrections

SDSS Filter Bands

SDSS images the sky through five broadband filters (u, g, r, i, z), spanning from near-UV to near-IR.

Apparent Magnitudes

SDSS provides several magnitude definitions: `petroMag`, `modelMag`, `cmodelMag`, `expMag`, `deVMag`. Each measures flux within a different aperture or profile fit.

The Petrosian Radius

Several of the magnitude definitions above reference the *Petrosian* radius. Unlike model-based radii (which depend on fitting a profile), the Petrosian radius is defined purely from the observed surface brightness profile. It is the radius R_P at which the ratio η of the local surface brightness to the mean enclosed surface brightness drops below a threshold:

$$\eta(R) = \frac{\bar{\Sigma}(0.8R, 1.25R)}{\bar{\Sigma}(0, R)} = 0.2$$

The Petrosian flux is then measured within $2R_P$, capturing a fixed fraction of the total light regardless of distance (in principle). This avoids the need to extrapolate a model profile to infinity. The `petroR50` and `petroR90` fields give the radii enclosing 50% and 90% of the Petrosian flux, respectively. Their ratio defines the *concentration* parameter, $c = R_{90}/R_{50}$, which we will use later as a morphological indicator.

Dust Extinction

The Milky Way dust extinction for each galaxy and filter band is provided in the SDSS dataset. We must correct for this to get intrinsic magnitudes.

K-corrections

To compare galaxies at different redshifts, we need K-corrections to account for the redshifting of the spectrum through the filter bandpass. We use the fitting functions of Chilingarian et al. (2010).

Galaxy Spectra and Color

To build physical intuition for why K-corrections matter, let's look at actual galaxy spectra alongside the SDSS filter bands. The spectra come from SDSS fibers (3" diameter) pointed at the galaxy centers. As stellar populations age, their spectra shift from blue to red and the 4000 Angstrom break becomes more prominent. When these spectra are redshifted, different parts of the spectral energy distribution (SED) pass through the filter bands, making K-corrections essential for accurate color measurements.

In-Class Exercise: Absolute Magnitudes

Goal

Convert apparent magnitudes to absolute magnitudes by applying the distance modulus and dust extinction corrections.

Task

Using the SDSS galaxy sample:

1. Compute the distance modulus $DM = 5 \log_{10}(d_L/10 \text{ pc})$ using Colossus.
2. Compute absolute magnitudes: $M_r = m_r - A_r - DM$, where A_r is the extinction.
3. Plot the histogram of M_r .
4. Print the median and the 0.5th/99.5th percentile range.

► Solution

Surface Brightness Profiles

Sersic Profiles

The Sersic profile is a widely used parametric form for galaxy surface brightness:

$$\Sigma(R) = \Sigma_e \exp \left[-b_n \left(\left(\frac{R}{R_e} \right)^{1/n} - 1 \right) \right]$$

where n is the Sersic index, R_e is the half-light (effective) radius, and $b_n \approx 1.9992n - 0.3271$.

Special cases:

- $n = 1$: exponential profile (disk galaxies)
- $n = 4$: de Vaucouleurs profile (elliptical galaxies)

In-Class Exercise: Sersic Profiles

Goal

Plot the Sersic surface brightness profile family to build intuition for how the index n controls the shape.

Task

For $n = 0.5, 1, 2, 4, 10$:

1. Compute $b_n = 1.9992n - 0.3271$.
2. Compute $\Sigma/\Sigma_e = \exp[-b_n((R/R_e)^{1/n} - 1)]$ on a log-spaced radial grid.
3. Also compute the surface brightness $\mu - \mu_e = 1.086 b_n((R/R_e)^{1/n} - 1)$ on a linear grid.
4. Make a two-panel plot: log-log (Σ/Σ_e vs R/R_e) and linear ($\mu - \mu_e$ vs R/R_e).

► Solution

SDSS Profile Fits

The SDSS pipeline fits both exponential ($n = 1$) and de Vaucouleurs ($n = 4$) profiles to every galaxy. The `modelMag` uses whichever fit is better.

Observed Radial Profiles

The SDSS pipeline measures azimuthally-averaged surface brightness profiles in circular annuli for each galaxy. Let's load these profiles and see how well the theoretical predictions hold: exponential galaxies should appear linear in R - μ space, while de Vaucouleurs galaxies should appear linear in $R^{1/4}$ - μ space. Profiles are shown in all five SDSS bands (u, g, r, i, z).

Galaxy Sizes

The effective radius R_e (half-light radius) varies enormously across the galaxy population. Let's look at the distribution in physical units (kpc).

Low Surface Brightness (LSB) Galaxies

Not all galaxies are readily detectable. Some spread their light over such a large area that their surface brightness falls below the sky noise, even though their total luminosity may be substantial. These **low surface brightness (LSB) galaxies** are systematically missed by magnitude-limited surveys like SDSS. The mean surface brightness within the half-light radius is

$$\langle \mu \rangle_{50} = m + 2.5 \log_{10}(2\pi R_e^2)$$

where m is the apparent magnitude and R_e is in arcseconds. Galaxies with $\langle \mu \rangle_{50} \gtrsim 23 \text{ mag/arcsec}^2$ are considered LSB. Recently, extremely diffuse objects called **ultra-diffuse galaxies (UDGs)** have been discovered in galaxy clusters, with $R_e > 1.5 \text{ kpc}$ but luminosities comparable to dwarf galaxies.

Size-Magnitude Relation

More luminous galaxies are larger. This is a fundamental scaling relation.

The Galaxy Luminosity Function

Malmquist Bias

A magnitude-limited survey is biased: brighter galaxies are visible to larger distances and are overrepresented. This is the Malmquist bias.

The V_{max} Method

To correct for Malmquist bias, we weight each galaxy by $1/V_{\text{max}}$, the inverse of the maximum volume within which it could have been observed given the survey's magnitude limit.

Corrected Magnitude and Color Distributions

The V_{max} correction dramatically changes the magnitude and color distributions. The corrected color distribution reveals a clear bimodality between blue and red galaxies.

In-Class Exercise: Schechter Luminosity Function

Goal

Construct the galaxy luminosity function from the SDSS data using the $V_{\max}$ method and fit a Schechter function.

Background

The Schechter luminosity function is

$$\phi(L) d\log L = \ln(10) \phi^* \left(\frac{L}{L^*} \right)^{\alpha+1} \exp\left(-\frac{L}{L^*}\right) d\log L$$

Task

1. Convert absolute magnitudes to luminosities using $\log_{10}(L/L_{\odot}) = 0.4 \times (M_{\odot,r} - M_r)$.
2. Build the luminosity function by histogramming $\log L$ weighted by $1/V_{\max}$.
3. Define a Schechter function and fit it to the data.
4. Plot both the corrected and uncorrected luminosity functions, plus the fit.

► Solution

Galaxy Morphology

The Color-Magnitude Diagram

Galaxy morphology correlates strongly with color: ellipticals tend to be red and concentrated, while spirals tend to be blue with lower concentration. The color bimodality is one of the most striking features of the galaxy population.

Morphology and Color

Galaxy images sorted by magnitude and color reveal the morphological trends: red, concentrated ellipticals vs. blue, extended spirals.

Concentration

Concentration $c = R_{90\%}/R_{50\%}$ measures how centrally concentrated a galaxy's light is. It separates spirals (low c) from ellipticals (high c) better than surface brightness alone.

Concentration and Color

The bimodality in color persists in the concentration-color plane: elliptical galaxies tend to be BOTH red and concentrated, while spirals tend to be BOTH blue and low-concentration.

Axis Ratio

Edge-on disk galaxies can mimic high-concentration ellipticals because the line-of-sight path through the disk is long, making the central surface brightness appear higher. The axis ratio b/a (minor-to-major axis from the best-fit profile) helps break this degeneracy: edge-on spirals have low b/a while true ellipticals tend to have $b/a \gtrsim 0.5$. A cut on high concentration *and* high b/a isolates genuine ellipticals more cleanly.

Machine-Learning Morphological Classifications

The simple diagnostics above (color, concentration, axis ratio) provide crude morphological separation but are far from perfect. [Huertas-Company et al. \(2011\)](#) applied machine-learning techniques to the SDSS sample, providing probabilistic classifications into four Hubble types: elliptical (E), lenticular (S0), early spiral (Sab), and late spiral (Scd). These probabilities are included in the UPenn photometric catalog of [Meert et al. \(2015\)](#). Let's visualize galaxies that are confidently classified into each type.

The Milky Way Rotation Curve

The rotation curve measures circular velocity as a function of radius. For the Milky Way, the disk + bulge model matches observations at small radii but falls far short at large radii. This is one of the strongest motivations for dark matter.